α-$BaCu_2Se_2$ Thin-Film XRD Analysis and Bayesian Optimization

NOMAD-driven data extraction, phase matching, and GP-based synthesis planning

1. What was done

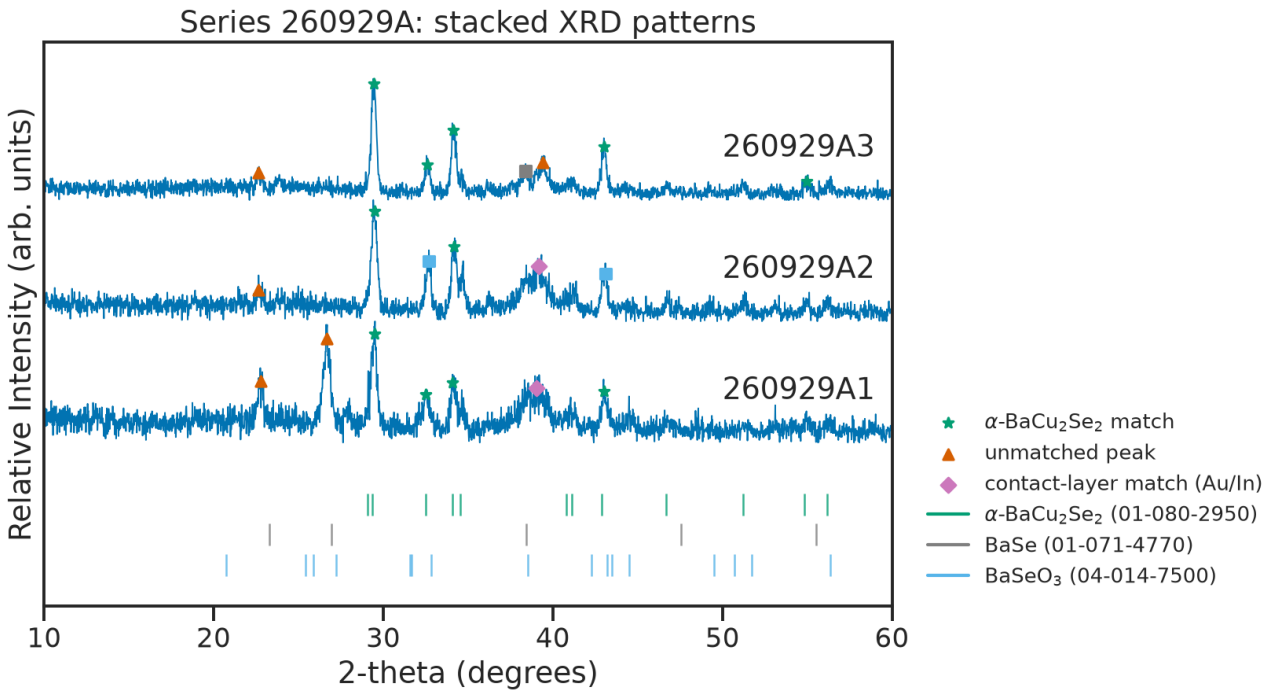
Synthesis records and XRD scans were pulled directly from the NOMAD Oasis REST API for the 42 α-$BaCu_2Se_2$ thin-film samples (INLThinFilmStack entries). Synthesis parameters (spin coating, soft/hard bake, selenization) and selenium-pulse settings were extracted for all samples; 35 include pulse on/off data. Each record also carries the precursor Cu/(Ba+Cu) ratio, the number of spin-coating cycles (deposited layers, excluding dynamic-dispense/seed-layer preparation steps), the Mo-coating flag, and the spin-to-selenization day gap. For the 28 films with XRD scans, diffraction patterns were matched against reference cards for the α-$BaCu_2Se_2$ target (01-080-2950) and the impurities β-$BaCu_2Se_2$ (04-008-2861), Cu_2Se (01-080-4377), CuSe (00-049-1456), BaSe (01-071-4770), $BaSe_2$ (04-004-8560), $BaSe_3$ (04-004-8559), $BaSeO_3$ (04-014-7500) and $BaSeO_4$ (04-011-7418); the α target always wins an overlap, otherwise the closest reflection decides. Au (00-004-0784) and In (00-005-0642) are classified separately as contact-layer matches with one shared plot symbol; their detected-peak intervals are excluded from phase-purity accounting and the unmatched/overlap penalty. In matching is restricted to series 260929A. Phase purity and crystallinity (1/FWHM of the strongest target match) were computed per film. A Gaussian-process Bayesian optimizer was trained on the 24 complete films to recommend the next synthesis conditions, optimizing objective = (0.7 × phase purity + 0.3 × normalized crystallinity) × (1 – 0.25 × A) over a five-input space: hard-bake temperature, selenization temperature and time, Cu/(Ba+Cu) ratio, and continuous Se pulse duty cycle. The on+off period is fixed at 3200 ms (3.2 s), so pulse frequency is fixed at 0.3125 Hz and is reported as metadata, not optimized. Intermediate duty cycles are model interpolations from three observed levels and should be treated as exploratory; recommended on-times are quantized to 1 ms increments. A long-form CSV of classified detected peaks is exported for sample-level audit. CBC is defined as $n_{Cu}/(n_{Ba}+n_{Cu})$; the candidate grid is 0.63–0.71, so a ratio of 1 (no Ba) is excluded. A is the unmatched/overlap share of detected peak-interval area, a screening proxy—not a quantitative phase fraction or a confirmed assignment of the ~23° feature.

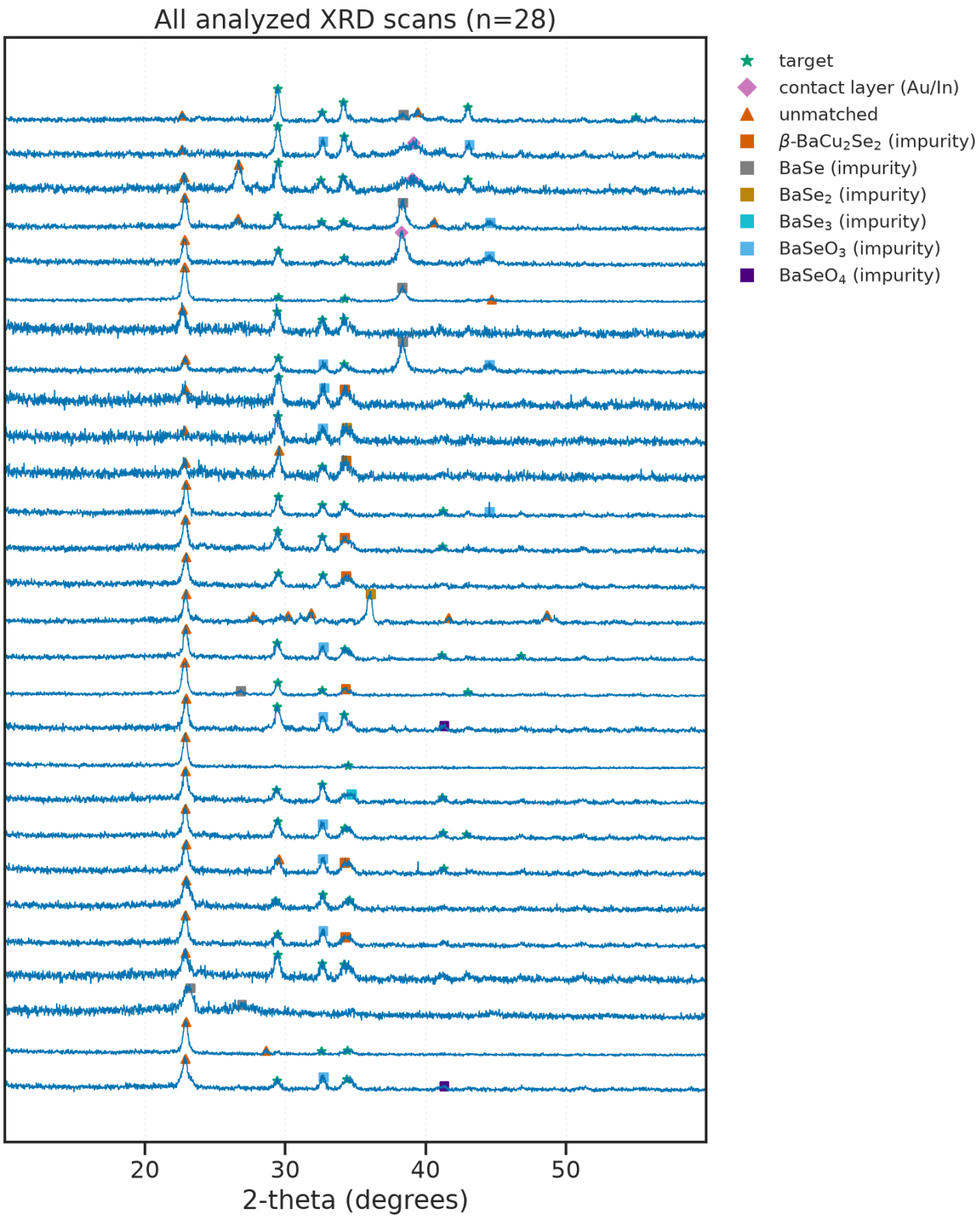
Refinement-based optimization readiness

The four-phase Le Bail export has 28 scans, 28 profile residuals and 26 converged fits. These are exploratory Bragg–Brentano proxy fits of GIXRD, not validated quantitative phase analysis. Le Bail intensities cannot establish alpha weight fractions, and the present fits do not provide calibrated phase-specific sizes or microstrain. The requested refinement-based BO is therefore blocked; the heuristic results below remain unchanged and must not be interpreted as physical weight fractions or crystallite sizes. Rwp is a profile residual, not an impurity fraction or proof of a correct structural model.

2. XRD phase identification

Stacked XRD patterns per series with reference bars for the α target (always shown) and only those impurity phases matched in the series' patterns. Markers flag target (star), impurity (square, colored by phase), shared Au/In contact (diamond), and unmatched (triangle) peaks. Au/In reference-card bars are omitted.

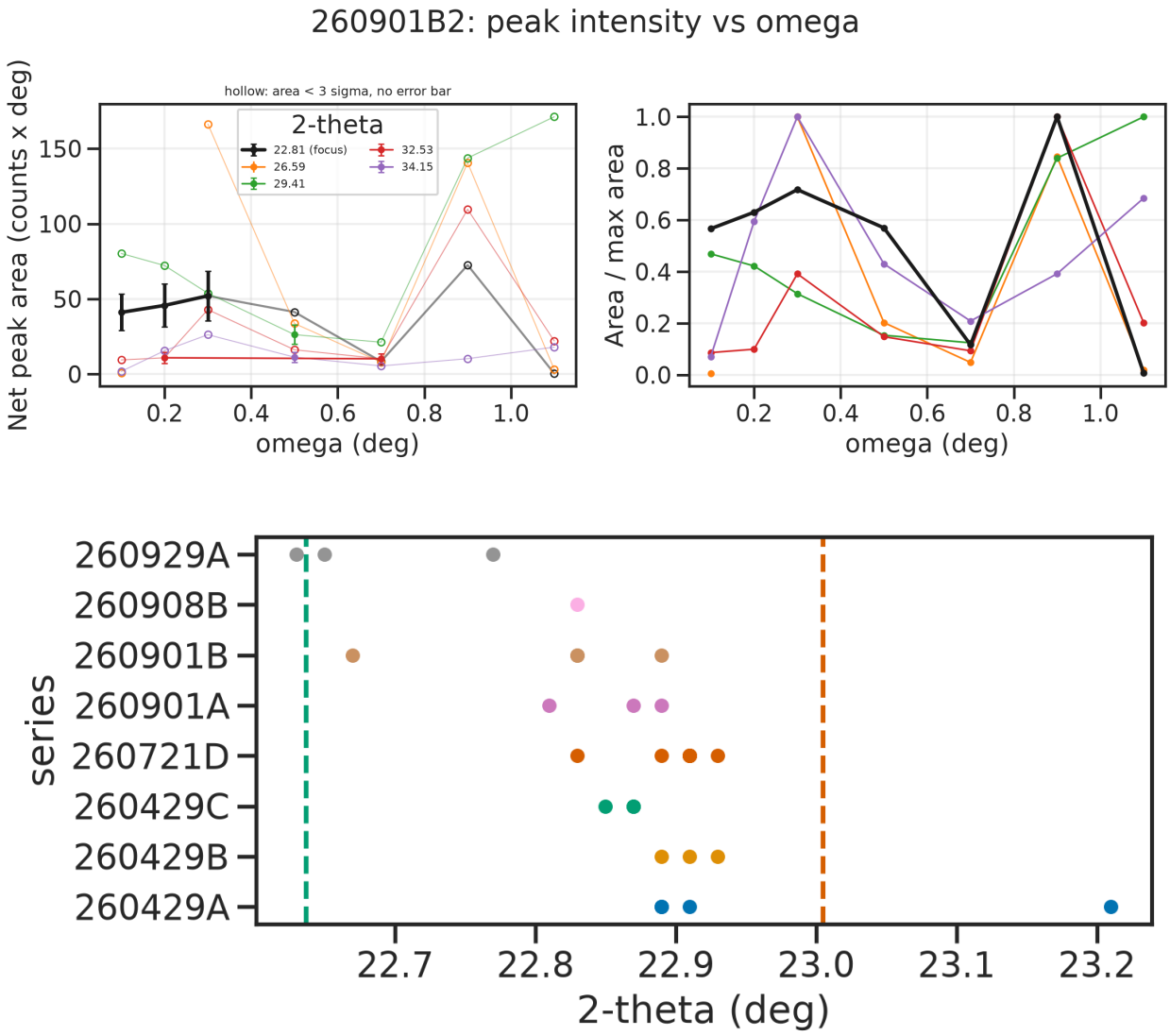

Series 260929A: stacked XRD patterns
Relative Intensity (arb. units)
260929A3
260929A2
260929A1
10
20
30
40
50
60
2-theta (degrees)
α-BaCu2Se2 match
unmatched peak
contact-layer match (Au/In)
α-BaCu2Se2 (01-080-2950)
BaSe (01-071-4770)
BaSeO3 (04-014-7500)



Omega study of sample 260901B2

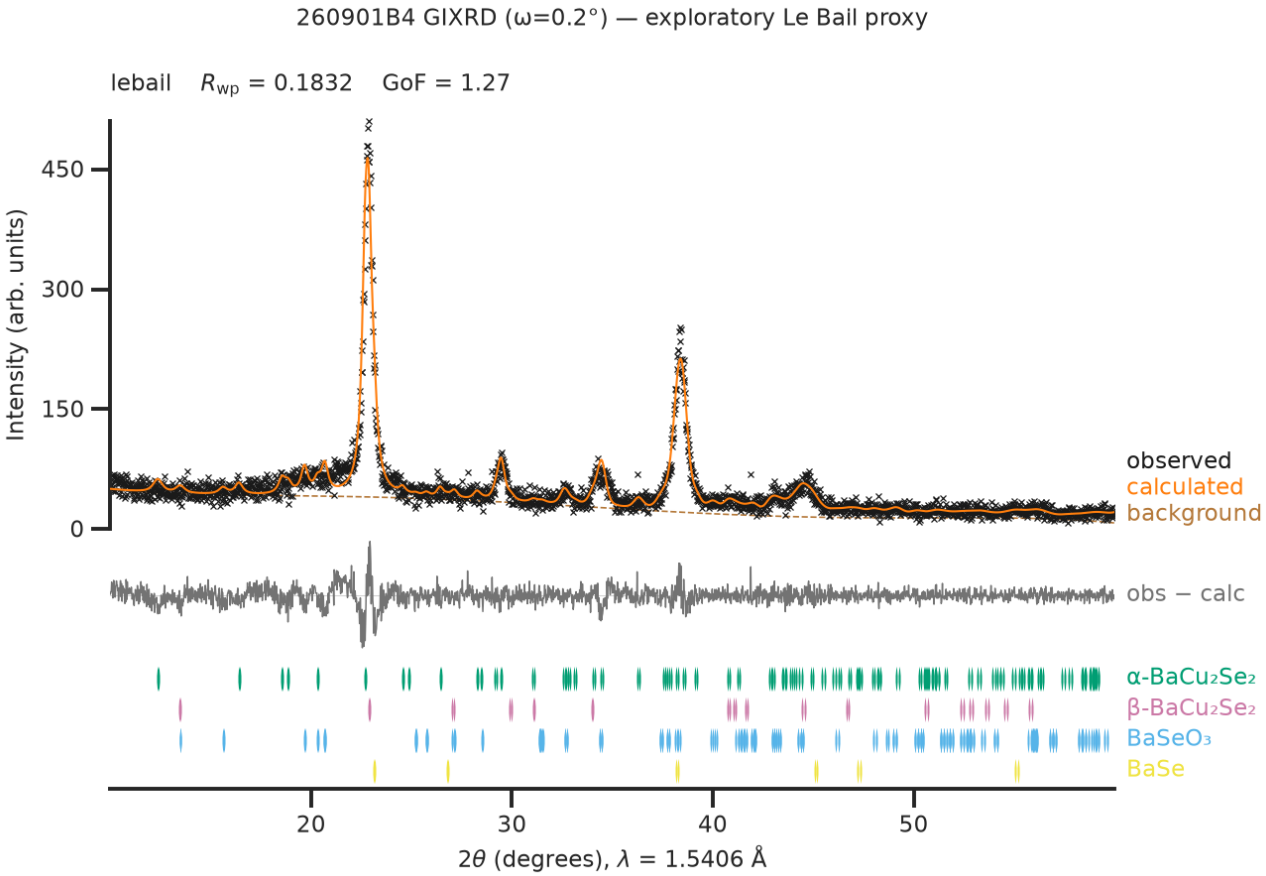
Seven grazing-incidence scans (omega 0.1–1.1°, 20–35° 2θ) were fitted peak by peak to test the origin of the ~23° feature (notebook 2.1; this sample only, not used by the optimizer). The feature sits at 22.81° 2θ with FWHM ≈ 0.4–0.5°. The 10 %-intensity matching used for purity does not assign it, but the cards list weak lines near it: α-$BaCu_2Se_2$ at 22.64° (4.6 %) and β-$BaCu_2Se_2$ at 23.01° (3.0 %), 0.17° below and 0.20° above the sample 260901B2 peak. Across all films the feature sits at 22.86–22.98° in series 260429A/B/C, 260721D and 260901A (closer to the β line) and at 22.68° in series 260929A (closer to the

α line), but proximity alone cannot establish the phase. The researcher reports that the SLG substrate has no sharp peak at 23°. Relative intensities differ from powder cards; texture is one possible explanation, but background, overlapping peaks and geometry also matter. A feature near 46.6–46.9° is a possible higher-order candidate, not an established second order: the analysis selected a maximum in a preset window without demonstrating a statistically significant peak or a common origin. Longer scans, calibration, rocking curves and pole figures can help distinguish the hypotheses.



An exploratory Le Bail profile fit (Bragg–Brentano proxy, cells fixed) was run for all 28 scans. The standard set is α- and β-$BaCu_2Se_2$ plus $BaSeO_3$ and BaSe, the impurities matched in most films (Cu_2Se was never matched). It was compared with other phase sets (table). Extra phases lower the overall R_{wp} but hardly change the 22.1–23.1° window: the four-phase standard fits it as well as nine phases. β alone fits the 23° window slightly better than α alone (lower RMSE in 18 of 28 scans) but the whole pattern much worse, and α plus β fits the window best. Le Bail intensities are free and the cells fixed, so this indicates that both polymorph lines help, not which one produces the 23° peak.

Phase set	R_{wp}	GoF	23° window RMSE	|peak pos. err.| (°)	calc/obs height
5 phases (previous CIFs)	0.190	1.29	23.0	0.074	0.76
9 phases (all cards)	0.184	1.24	19.9	0.053	0.78
α, β, Cu2Se, BaSe	0.199	1.34	19.8	0.072	0.86
α, β, BaSeO3, BaSe (standard)	0.193	1.30	20.0	0.070	0.84
α only + BaSeO3, BaSe	0.196	1.32	25.1	0.095	0.82
β only + BaSeO3, BaSe	0.238	1.61	23.0	0.066	0.82



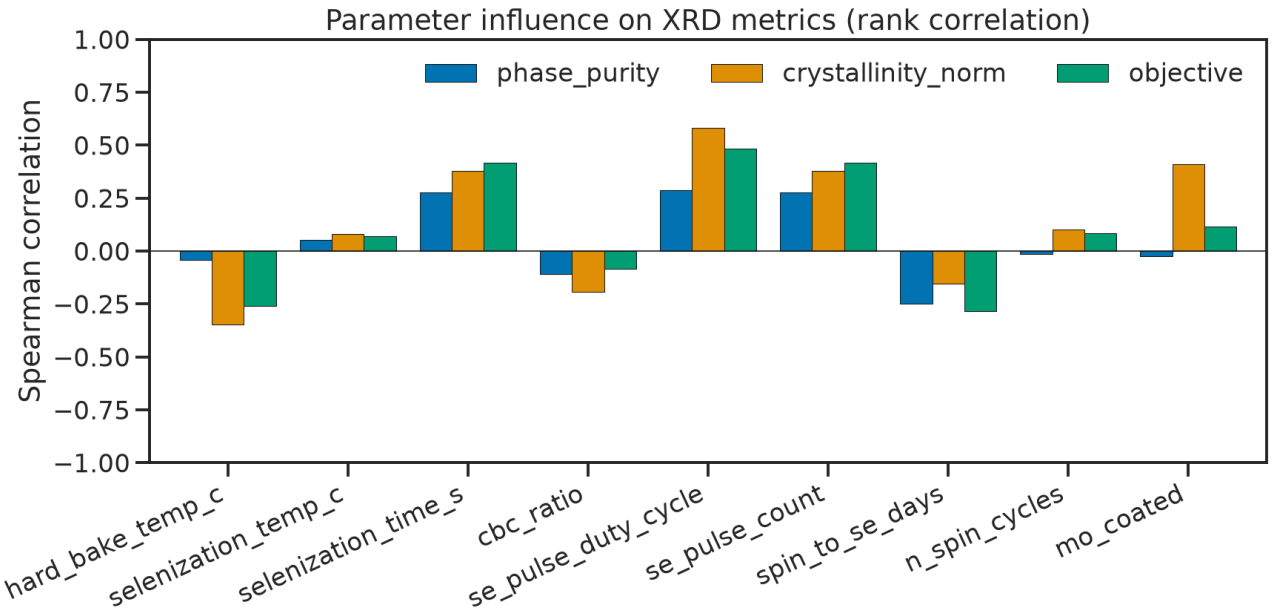
3. Main results

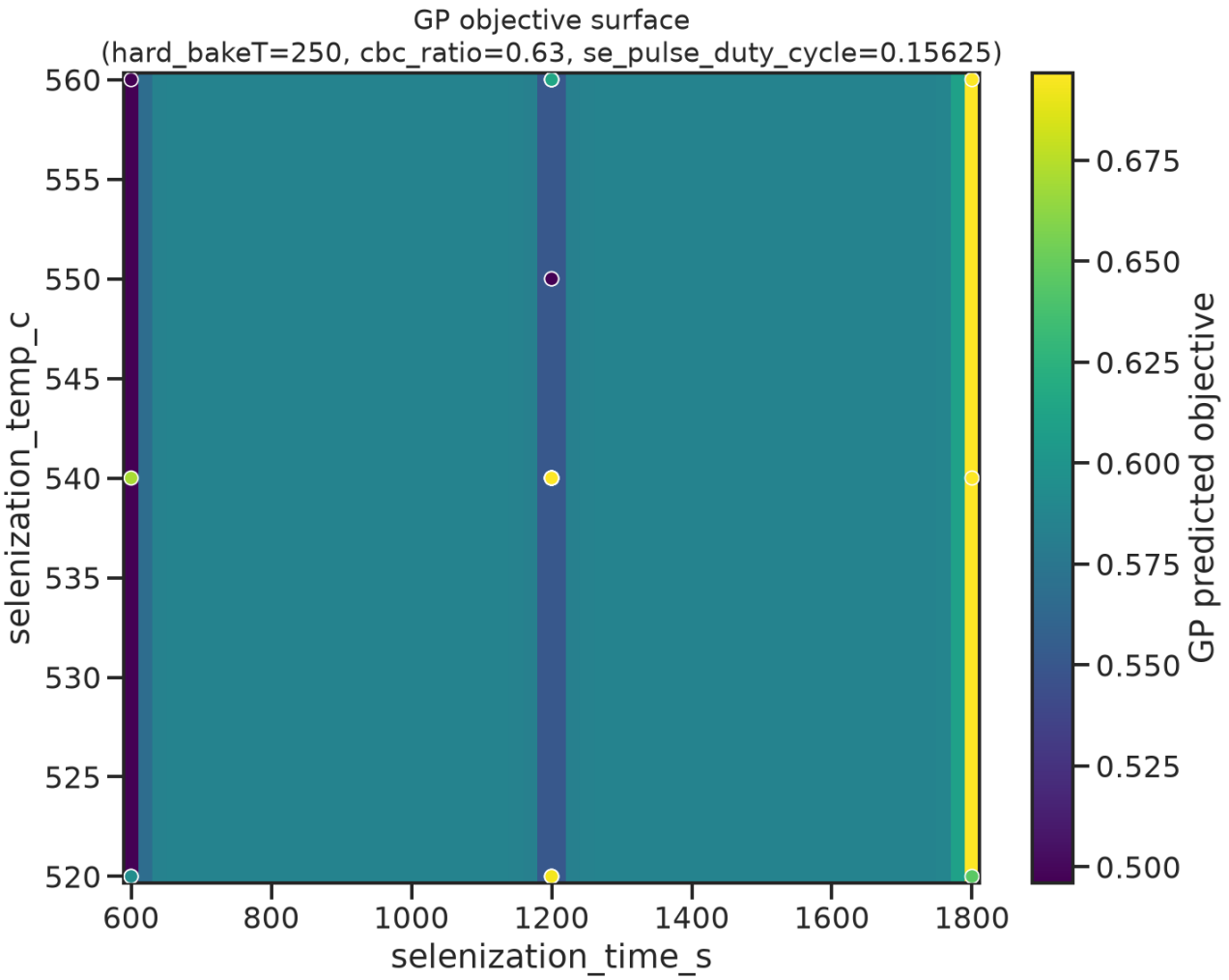
Across the analyzed films the mean phase purity is 0.68 and mean crystallinity is 2.16. The best film under the penalized score is 260429B2 (score 0.88; unpenalized score 1.00; purity 1.00, crystallinity 2.92, A 0.49). The optimizer uses the five selected inputs only. Other process and substrate descriptors remain available for descriptive exploration; duty cycle is modeled continuously, with the fixed pulse frequency excluded from the optimizer because it has no observed variation.

Purity–crystallinity weight sensitivity

The same films, penalty and search space were used for purity/crystallinity weights 0.5/0.5, 0.7/0.3 and 0.9/0.1. The table shows the best observed film and the first model-generated proposal in acquisition order, not a measured result. Observed score rank correlations versus 0.7/0.3 are 0.96 (0.5/0.5) and 0.93 (0.9/0.1). Recipe differences can also reflect acquisition randomness and uncertainty; seed 0 is held fixed.

Purity / crystal.	Best observed	First proposal: bake °C	Selen. °C / s	CBC	Se on / off ms
0.5 / 0.5	260429B2	225	530 / 1200	0.69	604 / 2596
0.7 / 0.3	260429B2	250	540 / 1800	0.68	500 / 2700
0.9 / 0.1	260901B2	250	560 / 1800	0.63	845 / 2355



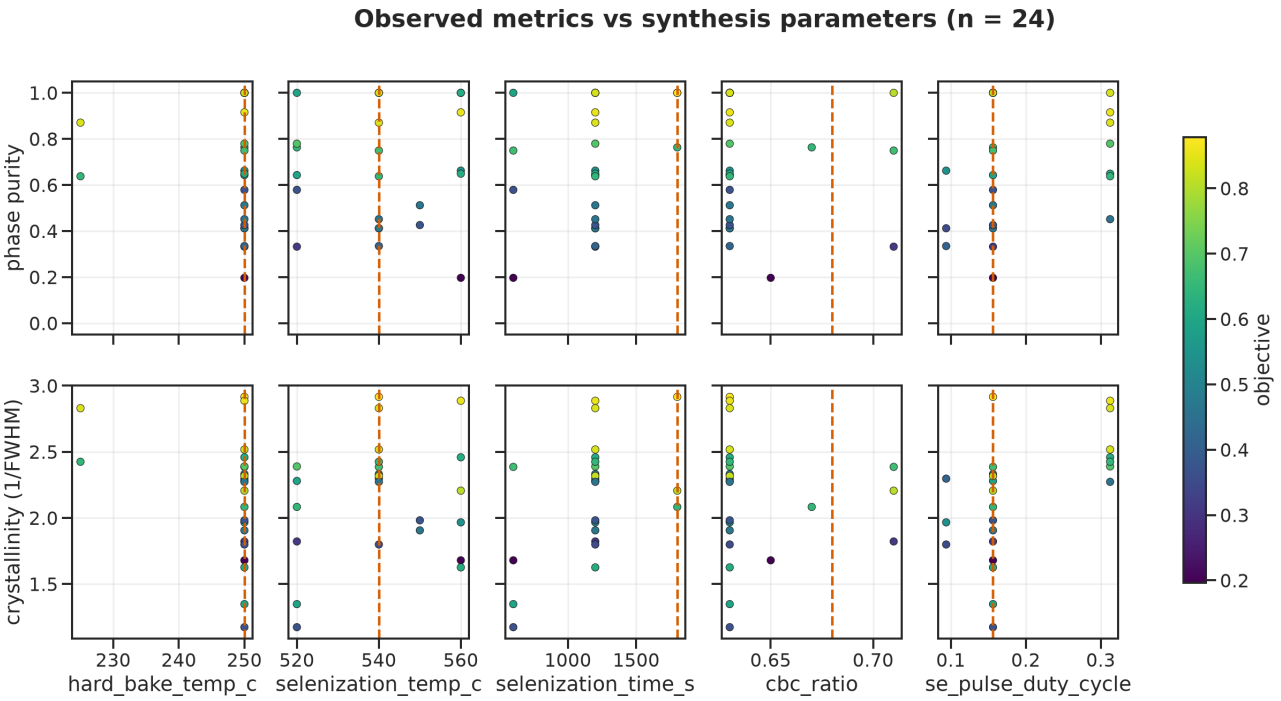


4. Suggested next experiments

Top Gaussian-process suggestions (predicted objective ± std), ranked by expected improvement:

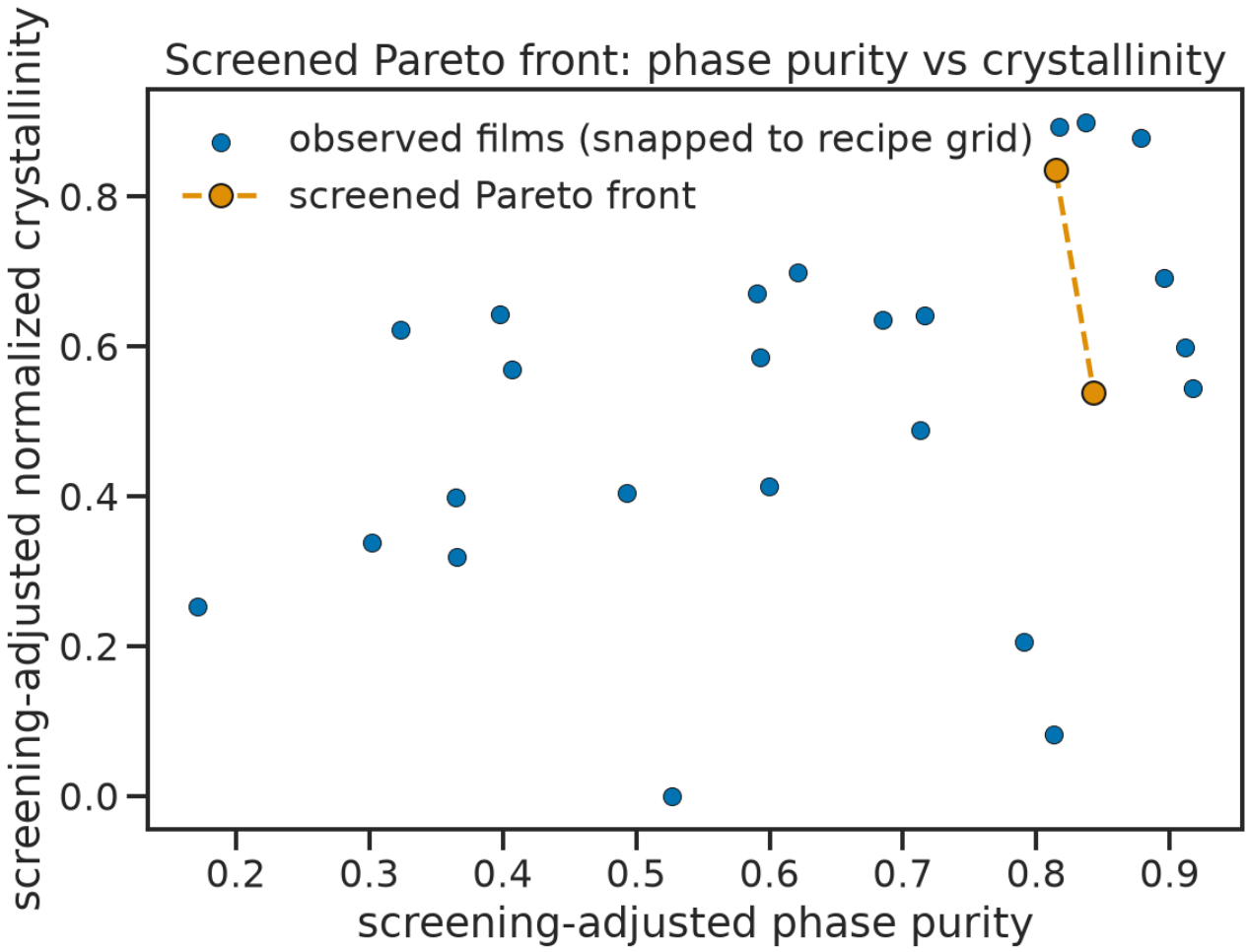
#	Hard bake °C	Selen. temp °C	Selen. time s	CBC ratio	Se duty cycle	Se on ms	Se off ms	Predicted score	GP std
1	250	540	1800	0.680	0.1562	500	2700	0.682	0.170
2	225	550	1800	0.630	0.3125	1000	2200	0.709	0.154
3	250	540	1800	0.660	0.2963	948	2252	0.662	0.179
4	225	540	1800	0.700	0.2931	938	2262	0.663	0.179
5	225	540	1800	0.690	0.2959	947	2253	0.662	0.179

Recommended first experiment: hard bake 250 °C, selenization 540 °C for 1800 s, CBC ratio 0.680, and Se pulse duty cycle 0.1562 (500 ms on / 2700 ms off) (predicted objective 0.68).



5. Multi-objective optimization (Ax)

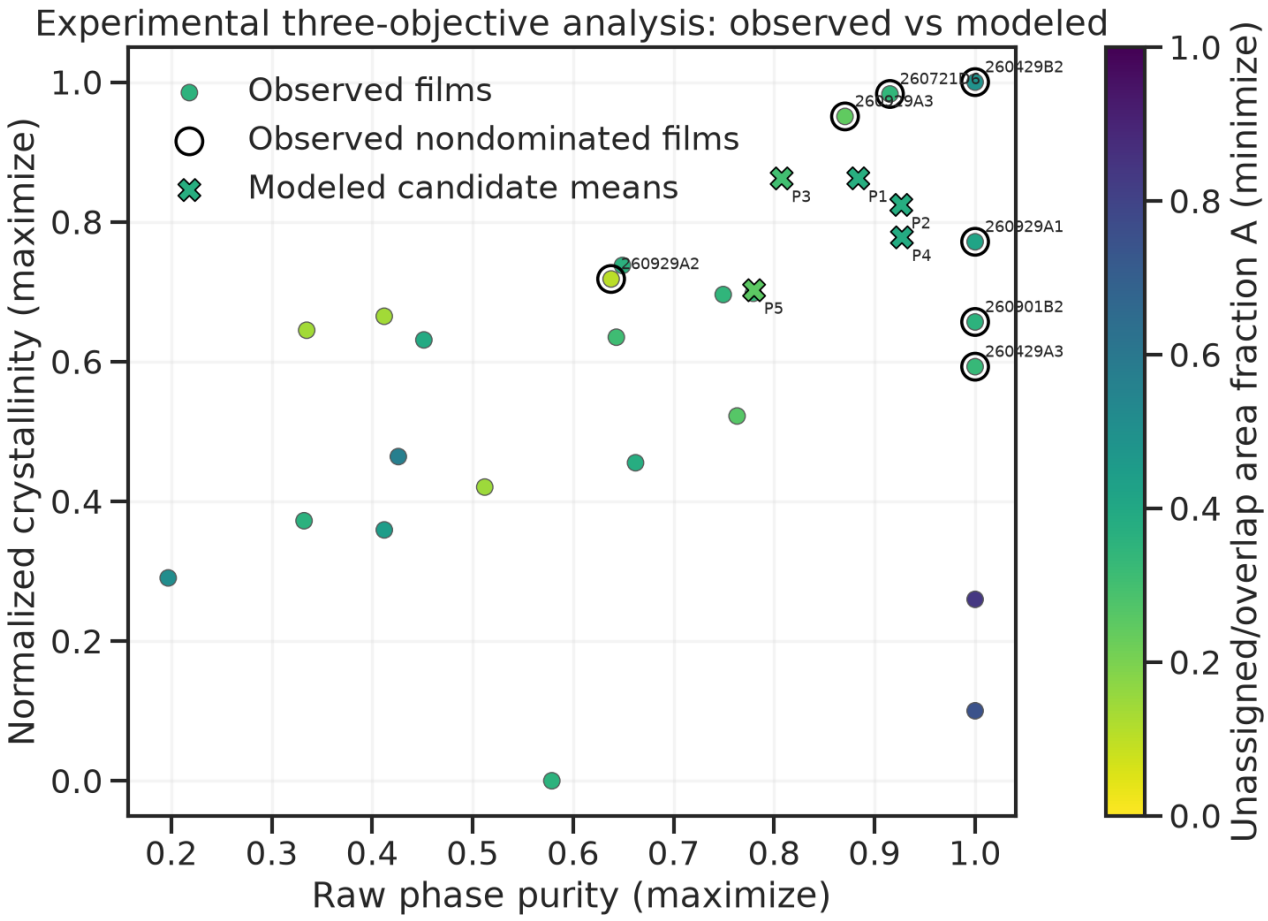
A complementary multi-objective search treats screening-adjusted phase purity and normalized crystallinity as two separate objectives: both are multiplied by the same (1 – 0.25 × A) factor. Ax (BoTorch) fits a Gaussian process per metric and finds the Pareto front via qNEHVI hypervolume maximization. Its five candidate inputs match notebook 3: hard-bake temperature, selenization temperature/time, CBC ratio, and continuous pulse duty cycle. Pulse period is fixed at 3.2 s (0.3125 Hz), so frequency is excluded as a constant. Duty-cycle suggestions are quantized to 1 ms on-time increments; interpolation between the three observed duty levels is exploratory. Numeric process recipes use the same explicit experiment-scale grids, including 25 °C bake steps. This remains a Pareto formulation, not the same scalar objective as notebook 3. A is an exploratory detected-peak-area screening proxy, not a phase fraction or a confirmed assignment of the ~23° feature.



#	Hard bake °C	Selen. temp °C	Selen. time s	CBC ratio	Se duty cycle	Se on ms	Se off ms
1	225	560	1500	0.630	0.3125	1000	2200
2	225	540	1800	0.630	0.1703	545	2655
3	250	560	1800	0.640	0.3125	1000	2200
4	250	530	1500	0.630	0.1897	607	2593
5	250	560	1800	0.630	0.0997	319	2881

Experimental three-objective comparison

An independent Ax experiment maximizes raw purity and normalized crystallinity while minimizing A directly, without multiplying the first two outcomes by the penalty. Reference thresholds (0, 0, 1.05) define the hypervolume baseline, not feasibility constraints. 7 of 24 complete films are nondominated among the observed films; the displayed model predictions are not measurements and do not establish an exact Pareto surface. Purity and A share peak classifications, and A includes the unassigned ~23° feature; it is not an impurity fraction. This comparison does not replace the production recommendations.


Experimental three-objective analysis: observed vs modeled
Observed films
Observed nondominated films
Modeled candidate means
260429B2
260721B2
260929A3
P3
P1
P2
P4
260929A1
260929A2
P5
260901B2
260429A3
Normalized crystallinity (maximize)
Raw phase purity (maximize)
Unassigned/overlap area fraction A (minimize)
0.0
0.2
0.4
0.6
0.8
1.0
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0